

Aswin Sanosh

Software Developer

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Portfolio: <https://aswinsanosh.vercel.app>

Professional Summary

Full Stack Developer and AI/ML Engineer with hands-on experience building web applications, real-time systems, and data-driven tools through freelance and startup work. Comfortable developing end-to-end products with Django, React.js, and modern backend frameworks, while also working on computer vision, unsupervised learning, and physics-informed AI models. Experienced in REST APIs, real-time data pipelines, and IoT integrations for projects such as stock prediction, flood forecasting, and smart transport systems. Also involved in mentoring software teams, improving system performance, and delivering practical solutions in research-focused and startup environments.

Core Skills

Programming Languages: Python, JavaScript, Java, C, C++, Typescript, PHP, R(Language), VB .NET, SQL, Dart, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework, React.js, Next.js, Node.js, Express.js, NestJS, Flutter, React Native, Three.js, Tailwind CSS

AI/ML & Data: NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv8, Hugging Face Transformers, MLflow

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase

Tools & Platforms: Git, VS Code, PyCharm, Google Colab, Figma, Blender, Arduino IDE

Domains: Full-Stack Development, AI/ML, IoT, Computer Vision

Professional Experiences

Software Mentor and Developer Feb 2025 – Present

Tessat Space Pvt. Ltd. (Startup), Kottayam, Kerala, India

- Mentoring software team and Developing full-stack applications and conducting AI/ML research.

Intern Dec 2025 – Mar 2026

National Chung Cheng University, Minxiong, Chiayi, Taiwan

- Digital twin design for flood inundation using physics-informed neural networks.

Web-Master Dec 2024 – Dec 2025

IEEE Computer Society Student Chapter, Saintgits College of Engineering

- Implemented web solutions for events, announcements, and engagement.

Intern July 2024 – Sep 2024

National Chung Cheng University, Taiwan

- Developed a stock prediction website using Django and React.js.

Teacher (Part Time) Feb 2024 – Dec 2025

Gurukulam Tuition Center, Kottayam, Kerala, India

- Taught Computer Science (C++, Website Development, Networking etc) for 12th standard students.
- Taught Physics, Chemistry and Mathematics for 10th standard CBSE students.

Chief Marketing Officer July 2023 – Feb 2024

IEDC@Saintgits, Kottayam, Kerala, India

Significant Projects

TrueSeal

Next.js, Tailwind CSS, Typescript, Django, AI, Celery, Redis, Redux

- Built an AI-powered recruitment platform integrating deep verification into ATS workflows to connect verified talent with data-driven businesses.

Digital Twin for Flood Inundation

Python, PyTorch, Graph Neural Networks, HEC-RAS

- Developed a physics-informed flood inundation digital twin using graph neural networks on unstructured meshes, integrating HEC-RAS hydrodynamic outputs with terrain/rainfall data and custom physical-constraint loss functions for stable, interpretable predictions.

Stock Prediction Website

React.js, Tailwind CSS Django, PostgreSQL

- Built a full-stack stock prediction platform with pattern detection, visual analytics, REST APIs, and real-time alerts.

Body Shape Classification

PHP, Python, OpenCV, OpenPose, Shapy AI

- Developed an AI-based body shape classification system using pose estimation, body geometry extraction, and anthropometric analysis for fit recommendations.

Ground Penetrating Radar (GPR) Image Classification

Python, Keras, DINOv3, DBSCAN, BIRCH, K-Means

- Combined feature extraction using DINOv3 and CNN backbones with clustering methods such as DBSCAN, BIRCH, and K-Means to improve pattern discovery and analysis quality.

Industrial Tool Classification

C++, OpenCV, HOG, Hu Moments

- Built a real-time industrial tool classification system using HOG and Hu Moments in OpenCV for manufacturing and inspection workflows.

Samyuktha 2025 Website

Next.js, Tailwind CSS Redux, Tailwind CSS, MySQL

- Built the official event platform for a college techno-cultural fest with registrations, announcements, participant dashboards, and coordination workflows.

CANSAT Flight Software

Django, Vite React.js, Tailwind CSS, MySQL, ESP32, Thermal Sensors, IoT

- Engineered flight software with telemetry dashboards, anomaly detection, and alerting workflows using ESP32-based microcontrollers and thermal sensor data for satellite operations.

Helekin Technologies Website

Next.js, Tailwind CSS, Three.js, MySQL

- Built a 3D-printing e-commerce platform with interactive model customization, order tracking, and product management workflows.

Redwills Interactive Website

Vite React.js, Tailwind CSS

- Developed an immersive studio website with interactive 3D visuals and a narrative-driven experience aligned with gaming aesthetics.

Education

Integrated Master of Computer Applications

2021 – 2026

Saintgits College of Engineering, Kottayam, Kerala, India

CGPA: 7.54/10

Certifications and Awards

- Developing Applications with SQL, Databases, and Django - Coursera
- Python for Machine Learning & Data Science - Udemy
- Electronics and PCB Design - Udemy
- Winner - Meta SparkAR Competition (April 2021)

References

Prof. Pao-Ann Hsuing, PhD, Dean of College of Engineering

Professor, Department of Computer Science & Information Engineering

Director, Taiwan-India Joint Research Center on Artificial Intelligence

Director, Research Center on Artificial Intelligence and Sustainability

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